Key Characteristics Associated with Diabetes

1. Data Quality & Pre-processing

- Scope: Analyzed 768 observations; focused on Outcome as the target.
- Cleaning: Replaced impossible "0" values (Glucose, BMI, Insulin) with NaN to eliminate statistical bias and ensure accuracy.

2. Bar Chart Analysis: Age-Risk Correlation

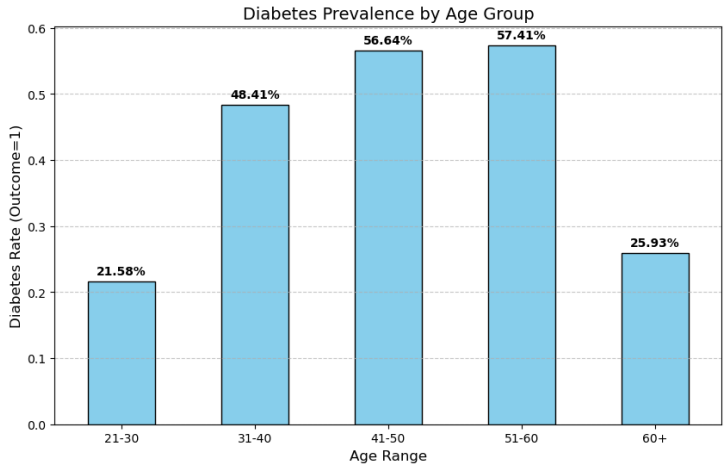
- Primary Trend: Prevalence follows a clear upward trend as age increases.
- Risk Peak: Highest risk is in the 41–60 bracket, peaking at 57.41% (nearly triple the 21–30 group).
- 60+ Group: Observed decline is due to sample sparsity (3.6% of data), not biological risk reduction.

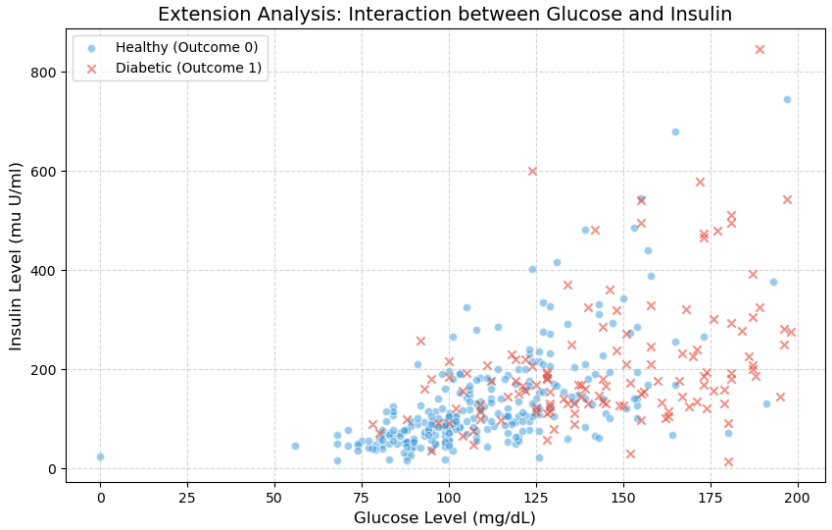
3. Statistical Robustness (Bootstrapping)

- Simulation: Performed B=1,000 iterations (n=30) to simulate the Glucose sampling distribution.
- Validation: Confirms a perfect Normal Distribution with a narrower Standard Error.
- Theory: Validates the sample mean as a reliable estimator based on the Central Limit Theorem.

4. Extension: Metabolic Interaction

- Integrity: Filtering "0" Insulin values provides a precise view of metabolic relationships.
- Risk Pivot: Identified a critical spike after age 40, where diabetes prevalence doubles.
- Interaction: Scatter plots show diabetic clusters at high-range Glucose and Insulin, indicating higher metabolic burden.





Summary:
Glucose is the primary driver, with diabetic averages **31.2 units higher** than healthy groups. Stratified data identifies a **critical "Risk Pivot" at age 40**, where prevalence spikes non-linearly. Validated by **B=1,000 bootstrap simulations**, this confirms **Age 40+** as the key window for intervention.